Problem A
Billboards
Time limit: 6 seconds



Billboards in the contest area

Each year, the ICPC (International Collegiate Programming Contest) has many sponsors. Since happy sponsors means happy contest, we plan to add a long billboard on one side of the contest area, so that our lovely sponsors can put their advertisements on it.

There are n wonderful sponsors in total, so to be fair, we would like to give each of them $1/n$ of the total area of the billboard. However, this is complicated by the fact that each marvelous sponsor potentially values sections of the billboard differently. For example, some sponsors want to put their ads near the entrance of the contest area, which may be at the corner where all contestants are guaranteed to see them when they enter. On the other hand, some sponsors may just want to put their ads in the middle, which is more likely to be broadcast in the live stream. (Don't ask us what happens if the entrance is at the middle! It's just an example.)

After talking with all of our delightful sponsors, the ICPC staff finds that the preference values of each sponsor can be modeled as a continuous piecewise linear function, where the domain of each of these value functions is the length of the billboard. Given this information, the ICPC staff wants to split the billboard into n sections, and give them to our n sponsors so that each magnificent sponsor gets at least $1/n$ of the value of the billboard from their point of view. That is, the area of the value function under the section that each sponsor gets must be at least $1/n$ of the total area of their own value function.

Figure A.1 shows a simple example. The billboard length is 10 and there are two sponsors, Orakle Software and Hal++, both of them having a value function with only one piece. Orakle's value function indicates that the company increasingly prefers sections of the billboard as you move left to right, while Hal++'s value function indicates the opposite. The area under each function is the total value of the billboard from each sponsor's perspective. The blue and the green areas shown are both larger than half of each total area, so cutting the billboard in the middle (the dashed line) is a valid solution, resulting in the billboard shown in Figure A.2. Note that there are many other divisions of the billboard that work as well (for example, cuttings at 4 or 6 are also valid solutions, if the sections of billboards are given to the correct sponsor).

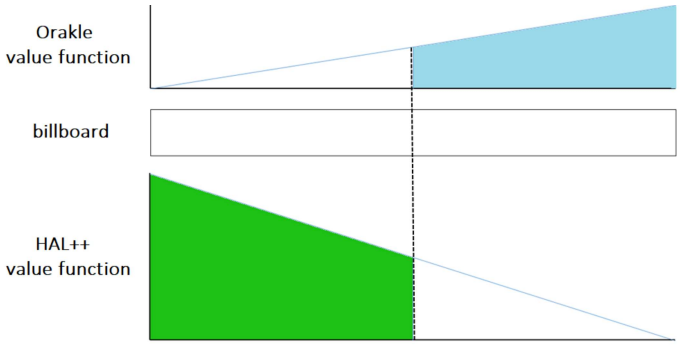


Figure A.1: Sample Input 1

Input

The first line contains two integers: n ($1 \leq n \leq 5\,000$), the number of glorious sponsors that the ICPC has (numbered 1 to n), and l ($1 \leq l \leq 10^6$), the length of the billboard.

Each of the remaining n lines starts with an integer m ($2 \leq m \leq 5\,000$), the number of points specifying this esteemed sponsor's value function. The remainder of the line contains m pairs of integers $(a_1, b_1), \ldots, (a_m, b_m)$ ($0 = a_1 < a_2 < \ldots < a_m = l$, $0 \leq b_i \leq 100$), where each pair represents an endpoint of one section of the piecewise linear function. The sum of all m's is at most 500 000 and it is guaranteed that at least one b_i is positive for each sponsor.



Figure A.2: Final Billboard corresponding to Sample Output 1

Output

Output n pairs of numbers (l_i, f_i) ($1 \leq i \leq n$), which indicate that the billboard section in the range $[l_{i-1}, l_i]$ is given to sponsor f_i (where $l_0 = 0$ is assumed, $l_0 < l_1 < l_2 < \ldots < l_n$, and l_n must be equal to l). The f_i values must be integers in the range $[1, n]$ where each integer appears exactly once, but the l_i can be real numbers. If there are multiple solutions, output any of them. If there is no solution, output `impossible`.

Note that you don't need to optimize anything else, as long as each sponsor gets at least $1/n$ of their total area. Each sponsor's allocated area may be below $1/n$ of their total area if the absolute or relative difference is at most 10^{-8}.

Sample Input 1	Sample Output 1
2 10 2 0 0 10 5 2 0 10 10 0	5 2 10 1

Sample Input 2	Sample Output 2
5 100 5 0 0 10 0 20 1 30 0 100 0 5 0 0 10 0 20 1 30 0 100 0 5 0 0 10 0 20 1 30 0 100 0 5 0 0 10 0 20 1 30 0 100 0 5 0 0 10 0 20 1 30 0 100 0	16.3245553203 1 18.9442719100 2 21.0557280900 3 23.6754446797 4 100 5

Problem B
Bingo for the Win!
Time limit: 1 second

Bingo is a game of chance for multiple players. Each player receives a sheet with some numbers, and a game master then calls out these numbers in a random order. Players cross off the numbers that they have heard, and the first player to cross off all their numbers wins the game. This basic version of the game has a reputation for being, well, a bit sedate. No particular action is required of the players except for not falling asleep.

In this problem we will analyze a more dynamic version of Bingo that requires quick thinking. In our version, called Speed Bingo, the game master also calls out the numbers from the sheets in a random order. However, whenever a number is called out, only the first player to signal that he or she has the number is allowed to cross it off their sheet. If a player has the same number multiple times, only one copy may be crossed off at a time. When multiple players have the same number(s) on their sheets, whoever has the fastest reaction time has an advantage in winning Speed Bingo. But how big an advantage? That's what we need your help to find out.

Formally, there are n players, each receiving a (possibly) different sheet of k (not necessarily distinct) numbers. Player 1 is faster to react than player 2, who in turn is faster than player 3, and so on, with player n being the slowest. Consider the following example, corresponding to the first sample input, where three players receive four numbers each:

1	2
3	4

Player 1

1	2
5	6

Player 2

3	4
7	8

Player 3

When number "1" is called for the first time, player 1—being faster—will get to cross it off their sheet. The second time "1" is called, player 2 will get to cross it off. So on average, we would expect player 1 to do better than players 2 and 3, since both of them will need some numbers that player 1 will get to first. However, since players 2 and 3 have no numbers in common, their performances will be independent of each other, even though player 2 is faster than player 3.

Suppose the game is played until all players have crossed off all of their numbers, that is, until all $n \cdot k$ numbers on all of the sheets (including appropriate repetitions) have been read. Assuming the order of the numbers is uniformly random, how likely is it for each player to finish last?

Input

The input describes a single game of Speed Bingo. The first line contains two integers n and k, the number of players and number of numbers on each sheet ($1 \leq n \leq 100$, $1 \leq k \leq 1\,000$). This is followed by n lines containing k integers each, where the ith line gives the numbers on the sheet for the ith player. All those numbers are between 1 and 10^9, inclusive.

Output

Output n lines, one for each player. The ith line should contain the probability that player i finishes last. All values must be accurate to an absolute error of at most 10^{-6}.

Sample Input 1	Sample Output 1
3 4 1 2 3 4 1 2 5 6 3 4 7 8	0.000000000 0.500000000 0.500000000

Sample Input 2	Sample Output 2
4 2 1 2 3 4 10 5 7 8	0.250000000 0.250000000 0.250000000 0.250000000

Problem C
Citizenship
Time limit: 4 seconds

It has been a long time since you moved to a different country and you have decided it is time to become a citizen. Your new country has a strict residency requirement for all applicants. To apply, you must have been physically present in the country for at least d days per year, for the past y consecutive years. These years are counted in 12-month periods backwards from the application date.

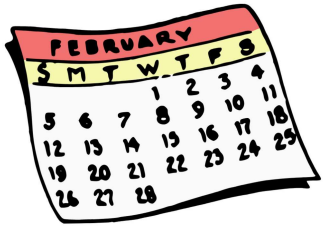


Image via Rawpixel, CC0

For this problem, assume that a calendar year has 12 months of 365 days, and each month has exactly the number of days below:

month	01	02	03	04	05	06	07	08	09	10	11	12
days	31	28	31	30	31	30	31	31	30	31	30	31

For example, if you were to apply on 2024–09–19 you must have been in the country for at least d days during the 12-month periods from 2023–09–19 to 2024–09–18, 2022–09–19 to 2023–09–18, and so on for y such periods.

You have lived in the country for at least y years, but having traveled a lot, you are not sure if you meet the residency requirement. Write a program that finds the earliest date you can submit your citizenship application given your travel history.

Input

The first line contains three integers n, y and d ($1 \leq n \leq 500$, $1 \leq y \leq 1\,000$, $1 \leq d \leq 365$). You have been out of the country n times and y and d specify the country's residency requirement as described above.

Each of the following n lines contains two dates in the form `YYYY-MM-DD` ($0000 \leq$ `YYYY` $\leq 5\,000$, $01 \leq$ `MM` ≤ 12, $01 \leq$ `DD` ≤ 31). You have been out of the country between the two dates, inclusive. All dates in the input are sorted in increasing order. The only dates which may be equal are dates on the same line. All given dates are valid.

Output

Output the earliest date on which you meet the residency requirement. The date must be after the last date of the input.

Sample Input 1	Sample Output 1
3 5 240 2022-02-28 2022-10-01 2022-11-11 2022-11-11 2023-12-30 2024-01-01	2024-05-31

Sample Input 2	Sample Output 2
3 5 240 2011-11-11 2012-12-12 2022-02-28 2022-10-01 2025-01-01 2025-06-30	2028-02-26

Problem D
Doubles Horseback Wrestling
Time limit: 4 seconds

The Nomadic Games Exploratory Committee (NGEC) is floating the idea of a doubles horseback wrestling tournament with pairs of riders astride single horses. They have advertised a pilot tournament, and n eager riders have signed up to compete! So now the NGEC needs to pair the riders in order to make the tournament both fair and exciting.

The Central Asian Audaryspak League (CAAL) maintains a list of all horseback wrestlers and their ratings. From their previous experience with ordinary horseback wrestling, the NGEC has decided that the pairs are best balanced if the ratings of their two riders add up to a particular integer, s.

For obscure licensing reasons, the CAAL refuses to release the exact rating of each rider. But the NGEC has some good estimates, knowing that any rider i's true rating r_i lies in an interval $[l_i, u_i]$. So the NGEC would consider pairing two riders i and j if there are ratings $r_i \in [l_i, u_i]$ and $r_j \in [l_j, u_j]$ such that $r_i + r_j = s$.

The NGEC wants to form as many non-intersecting pairs of riders as possible. You need to help them.



Horseback wrestling in Kyrgyzstan by Theklan
Wikimedia Commons, CC BY-SA 4.0

Input

The first line contains two integers n and s ($2 \le n \le 2 \cdot 10^5$, $1 \le s \le 10^9$), denoting the number of riders and the desired sum of ratings of riders in a pair. Riders are numbered 1 to n. This is followed by n lines, where the ith line contains two integers l_i and u_i ($1 \le l_i \le u_i \le 10^9$), denoting the rating range of the ith rider.

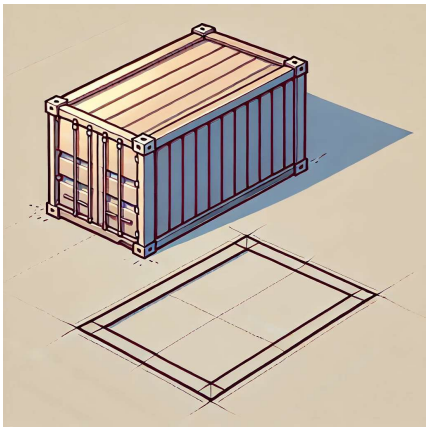
Output

Output k, the maximum number of riding pairs that can be formed, followed by k pairs of integers, denoting the numbers of the riders forming each pair. If there are multiple ways to pair off the riders, output any one.

Sample Input 1	Sample Output 1
6 10 6 7 1 4 2 2 3 8 5 7 9 9	2 6 2 3 4

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Problem E
Flipping Container
Time limit: 2 seconds



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A large, cuboid-shaped shipping container has been delivered to your shipping yard. Before you can open it up, you first need to move the container to a different location in your yard.

Normally, you would simply lift up the container using a crane and drop it in its target location. Unfortunately, your crane is broken, and the only way for you to move the container is by carefully flipping the container over one of its bottom four edges, resulting in a 90-degree rotation around the axis running along that edge. Your hope is that you will eventually be able to reach the target location by repeating this action enough times.

Just how many times will you need to perform this flip operation to get the container to the right spot? Note that the orientation of the container changes as you move it across the yard, but in the end it needs to be in the same orientation as at the start. Two orientations are considered the same when they have the same length along each of the three axes of the coordinate system. The three side lengths of the container are distinct.

Input

The first line contains three distinct integers a, b and c ($1 \leq a, b, c \leq 1\,000$), the dimensions of the container in meters. In the initial orientation of the container, the sides of length a run in east-west direction, the sides of length b run in north-south direction, and the sides of length c run in up-down direction.

The second line contains two integers x and y ($-10^{18} \leq x, y \leq 10^{18}$) giving the target location of the container. The container needs to be moved by x meters in the east-west direction and by y meters in the north-south direction (positive numbers indicate moving to the east/north, negative numbers indicate west/south).

Output

Output the least number of times the container needs to be flipped to reach its target location. If it is not possible to reach the target location, output `impossible`.

Sample Input 1	Sample Output 1
3 4 5 8 0	2

Sample Input 2	Sample Output 2
3 4 5 -8 9	4

Sample Input 3	Sample Output 3
3 4 5 123 45	40

Sample Input 4	Sample Output 4
20 10 30 13 37	impossible

Problem F
Friendly Rivalry
Time limit: 2 seconds

The leaders of the International Coalition for Planetary Change (ICPC), a non-profit fighting for environmental awareness, are worried that their regional chapters are not doing enough to make a real impact on climate change. Inspired by the latest studies that competition is one of the best motivators, they have decided to start a competition between their chapters.

At the same time, the ICPC does not want to slow the spread of ideas. To encourage chapters to share effective climate change methods, the ICPC has decided to assign their $2n$ chapters into two teams, the green team and the blue team. For balance, each team should consist of exactly n chapters.

To ensure the teams are not getting in each other's way, the ICPC wants the two teams to be as far apart as possible. Specifically, the Euclidean distance between the closest pair of chapters belonging to different teams should be as large as possible.

Help the ICPC set up the teams according to these rules.

Input

The first line contains an integer n ($1 \leq n \leq 500$), the size of each team. Chapters are numbered from 1 to $2n$. Each of the remaining $2n$ lines contains two integers x_i and y_i ($-10^9 \leq x_i, y_i \leq 10^9$), the Cartesian coordinates of the location of the ith chapter. All chapters are at distinct locations.

Output

Output $n + 1$ numbers. The first number is the distance between the two closest chapters belonging to different teams. The next n numbers are the chapters belonging to the blue team. If there are multiple ways to divide the teams with the same minimal distance, output any of them. The distance should have an absolute or relative error of at most 10^{-6}.

Sample Input 1	Sample Output 1
2 0 1 1 0 1 1 0 0	1.000000 1 2

Sample Input 2	Sample Output 2
2 0 1 -1 -1 1 0 2 2	2.236068 4 2

Sample Input 3	Sample Output 3
3 0 0 1 1 2 2 3 3 4 4 5 5	1.414214 1 2 3